MIL-E-7104D(WEP)
5 APRIL 1962
SUPERSEDING
MIL-E-7104C(WEP)
16 SEPTEMBER 1960

MILITARY SPECIFICATION

EJECTOR, RETRO, PNEUMATIC, MARINE MARKER A/A47U- 1 (*)

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall launch rounds from Naval ASW Aircraft in an aft direction with an ejection velocity equal to the aircraft speed so that the round will mark the spot in the water which was directly below the aircraft at the instant of launching the round.

1.2 Classification - The Marine Marker Pneumatic Retro-Ejector A/A47U-1(*) shall consist of the following items:

Items	Applicable Paragraph
Launcher LAK-26A/A47U-l	3.6.1
Local Control PEK-22/A47U-l	3.6.2
Remote Control PEK-23/A47U-l	3.6.3
Air Reservoir TMK-60/A47U-l	3.6.4
Magazine MAK-22A/A47U-l	3.6.5

2. APPLICABLE DOCUMENTS

2.1 General - The applicable documents, of the issue in effect on the date of invitation for bids, form a part of this document to the extent specified herein:

SPECIFICATIONS

Military

MIL-E-5272	Environmental Tesing, Aeronautical and Associated Equipment, General Specification for
MIL-P-5518	Pneumatic System, Design, Installation, and Test in Aircraft
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery

FSC 1055

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:07Z
Check the source to verify that this is the current version before use.

MIL-P-8564	Pneumatic Components, Aeronautical - General Specification for
MIL-P-8573	Reservoirs, Air, Non=Shatterable Steel
MIL-A-8591	Airborne Stores and Associated Suspension Equipment, General Destgn Criteria for
MIL-L-22769	Launcher, Weapons, Airborne, and Associated Equipment; General Specification for
MIL-R-23094	Reliability Assurance for Production Acceptance of Avionic Equipment, General Specification for

STANDARDS

Military

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-756	Reliability of Weapons Systems, Procedures for Prediction and Reporting Prediction of

PUBLICATIONS

Bureau of Naval Weapons

EI-403	Installation Instructions for the Ejector, Retro, Pneumatic, Marine Marker A/A47U- 1

2.2 Availability of Documents

(1) When requesting specifications, stan&rds, drawings, and publicatlone refer to both title and number. Copies of this specification and applicable specifications requested by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania

3. REQUIREMENTS

* 3.1 General - The equipment shall conform with all the applicable requirements of Specification MIL-L-22769 unless specified herein.

3.2 Performance - Unless otherwise specified, values set forth to establish requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations hom the performance under standard conditions will be specified.

* 3.2.1 Muzzle Velocity - The muzzle velocity shall be obtained when launching the Mark 25 Mod 0 Marine Location Marker.

3.2.1.1 Range - The range of muzzle velocity and Control shall be from 90 to 200 knots.

3.2.1.2 Tolerance - The muzzle velocity shall not vary more than ± 5% from the control. setting.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:07Z
Check the source to verify that this is the current version before use.

3.2.2 Cyclic Rate - A separate release pulse of 0.020 second minimum duration shall be required to eject each round and the equipment shall prepare itself automatically for the firing of the next round within 10 seconds.

3.3. Power Requirements

3.3.1 Electrical Power - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from a 28-V *DC*, Category "B", power sources with characteristics as defined in MIL-STD-704. The power required shall not exceed 3 amperes.

3.3.2 Pneumatic - The equipment shall meet all applicable requirements of Specifications MIL-P-8564 and MIL-P-5518 and shall give specified performance from 3000 psi sources in accordance with Specification MIL-P-5518.

* 3.4 Reliability - The equipment shall be capable *of the* operational stability, *mean*-time-between- failures, and total operating life specified herein. Equiprnent reliabilit *y* shallbe analyzed, calculated, and reported in accordance with MIL-STD-756.

* 3.4.1 Operational Stability - Each unit *of the* equipment (see 1. 2) shall operate with optimum performancer 100 hours, continuously or intermittently, without the necessity for readjustment *of* any parts which are inaccessible during flight. The equipment shall be capable of launching 250 rounds during this service life.

* 3.4.2 Total Operating Life - Each unit of the equipment (see 1.2) shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement within this interval and the life of such parts shall be specified by the manufacturer. The equipment shall be capable of launching 5000 rounds during the total operating life. Each part shall have a life of at least 1000 launchings.

* 3.4.3 Mean-time-between-failures - Each unit of the equipment (see 1.2) shall have a mean-time-between-failures of at least 200 hours and shall be capable of launching at least 500 rounds between failures.

* 3.4.4 Equipment Duty Cycle - The equipment duty cycle shall be 8 hours time-on and 16 hours time-off. During the time-on 20 romds shall be launched. The muzzle velocity control setting shall be changed for each launch. Performance characteristics (3.2) shall be measured for each launch and the muzzle velocity shall not vary more than ±3% from the control setting on the remote control which shall be used for all launches. The percentage of rounds launched at each velocity shall be approximately as follows:

Muzzle Velocity	Percentage of Rounds Launched
90	0.4
100	0.8
110	1.2
120	2.2
130	3.4
140	5.4
150	7.8
160	10.6
170	13.6
180	16.2
190	18.6
200	19.8

Source: https://assist.dla.mil - Downloaded: 2026-10-17T08:07Z
Check the source to verify that this is the current version before use.

3.4.5 Accept-Reject Criteria

* 3.4.5.1 Preproduction Tests - All sample units of the preproduction sample shall equal or exceed the operational stability, mean-time-between-failures, and total operating life requirements specified herein for acceptance.

* 3.4.5.2 Acceptance Tests Sampling Tests - Acceptance or rejection of lots represented by sampling test samples shall be made in accordance with Figure 12 of Specification MIL-R-23094.

3. 5 Design and Construction

3.5.1 Temperature - The temperature range shall be -32°C to +75°C

3.5.2 Acceleration - The equipment shall operate satisfactorily after being subjected to the design limit load and to design yield load acceleration specified in Specification MIL-A-8591; shall operate satisfactorily under positive design limit load accelerations; and shall not suffer structural failure when subjected to design ultimate load accelerations. Operatlon *of the* equipment is not required after being subjected to loads in excess of design yield load. (Note: The diagrams tn Specification MIL-A-8591 show the direction of the load; the accelerations *are* opposite. Upward and aft acceleration are positive.)

3.6 Detail Requirements

3.6.1 Launcher LAK-26A/A47U-l

3.6.1.1 Function - The launcher shall feed rounds from the magazine into the launching tube and eject the rounds at the muzzle velocity set on either the remote or local control, as selected.

* 3.6.1.2 Form Factor - Thee envelope dimensions of the launcher shall not exceed 70. inches long by 10 inches wide by 18 inches high in accordance with EI-403.

* 3.6.1.3 Weight - The weight of the launcher shall not exceed 70 pounds.

3.6.1.4 Contents - The launcher shall contain the following assemblies, subassemblies, circuits, etc.

3.6.1.4.1 Mounts - The launcher shall have a muzzle mount and a breech mount in accordance with EI-403.

3.6.1.4.2 Tube - The launcher shall have *a launching tube* having an inside diameter of 3.000 ± 0.005 inches.

3.6.1.4.3 Feed Mechanism - The Launcher shall have a feed mechanism for transferring rounds from the magazine to the launching tube.

3.6.1.4.4 Storage Chamber - The storage chamber shall contain the pressurized air charge used to propel the rounds and shall have a valve mechanism to release the air charge behind the rounds.

3.6.1.4.5 Charging Valve - The charging valve admit pressurized air to the storage chamber when actuated by the local or remote control, *as* selected.

3.6.1.4.6 Bleed Valve - The bleed valve shall *release* pressurized air from the storage chamber at a controlled rate when actuated by the local *or* remote control, as selected, or when the equipment is turned "OFF".

Source: https://assist.dla.mil -- Downloaded: 2026l0-17T08:07Z
Check the source to verify that this is the current version before use.

3.6.1.4.7 Round-in-place Switch - The round-in-place switch, in conjunction with other parts of the circuit shall provide a signal to operate the "ready" indicator on the remote control

3.6.1.4.8 In-battery Switch - The in-battery switch, in conjunction with other parts of the circuit, shall provide a signal to operate the "ready" indicator on the remote control.

3.6.1.4.9 Rounds Counter - The rounds counter shall record the total number of rounds ejected from the launcher.

* 3.6.1.4.l0 Tube Obstruction Piston -At all times when the storage chamber is depressurized, the tube obstruction piston shall prevent passage of a round through the tube beyond the in-battery position. Pressurization of a storage chamber shall withdraw the obstruction. Launching of rounds shall be prevented until the obstruction is fully withdrawn from the tube.

* 3.6.1.5 Electrical Connection - Connections to external circuits shall be provided by an MS3102A-18-1P connector as follows:

Pin Description	Function
A	Ground
B	Spare
C	Ready indication
D	Spare
E	Spare
F	Load
G	Bleed
H	Release
I	Swing and feed
J	Spare

3.6.2 Local Control PEK-22/A47U-l

3.6.2.1 Function - The unit shall provide for local control of the launcher.

3.6.2.2 Form Factor - The envelope dimensions of the local control shall be in accordance with EI-403.

* 3.6.2.3 Weight - The weight of the local control Shall not exced 3 pounds.

3.6.2.4 Contents - The local control shall contain the following assemblies, subassemblies, clrcuits, etc.

3.6.2.4.1 Mounting - The mounting shall be in accordance with EI-403.

* 3.6.2.4.2 Face Plate - The face plate shall mount devices necessary for operation of the equipment and shall have suitable legends engraved thereon.

3.6.2.4.3 Chassis - The chassis shall have provisons for mounting the parts which cannot be attached to the face of the unit.

Source: https://assist.dla.mil -- Downloaded: 202610-17T08:07Z
Check the source to verify that this is the current version before use.

* 3.6.2.4.4 Local-Remote Switch - The local-remote switch shall be a two position, guarded switch. In the remote position the switch shall interrupt electrical power from the aircraft to the local control but it shall not interrupt the circuits from the remote control. In the local position the switch shall interrupt the circuits from the remote control and shall apply aircraft power to the local control.

* 3.6.2.4.5 Cover Box - The cover box shall adequately shield the equipment from external interfering fields, shall adequately confine the internal fields, and shall provide required mechanical protection during storage, handling, and installation. The cover box shall be attached to the body of the control panel by means of quarter-turn quick fasteners *at a* minimum number of points. Complete removal of the cover box from the control panel shall be possible without disassembling the control panel in any other respect. The cover box shall have a cut-out so that the name plate will be easily visible with the cover box in place *on the* control panel.

3.6.2.4.6 Charge Switch - The charge switch shall be a push-button type switch. Pressing the switch shall apply power from the power switch to the charging valve allowing pressurized air to *enter the* storage *chamber.* Releasing the switch shall interrupt the power to the charging valve interrupting the flow of pressurized air to the storage chamber.

3.6.2.4.7 Bleed Switch - The bleed switch shall be a push-button type switch, Pressing the switch shall interrupt the electrical power from the power switch to the bleed valve allowing the pressurized air in the Storage chamber to flow out. Releasing the switch shall apply power from the power switch to the bleed valve interrupting the flow of air out of the storage chamber.

3.6.2.4.8 Release Switch - The release switch shall be a guarded push-button type switch Pressing the switch shall apply electrical power from the power switch to the firing valve allowing the pressurized air in the storage chamber to eject the round in the launching tube.

3.6.2.4.9 Air Reservoir Gauge - The air reservoir gauge shall be a 0 to 5000 psi pneumatic pressure gauge which shall indicate in psi the pneumatic pressure in the air reservoir. Provisions shall be made for lighting the gauge either with integral lighting or with a lighting shield.

3.6.2.4.10 Ground Speed Gauge - The ground speed gauge shall indicate the pneumatic pressure in the storage chamber and shall be calibrated in knots of ground speed.

* 3.6.2.5 Electrical Connection - Connections to external circuits shall be provided as follows:

Receptacle Type	Pin Designation	Function
MS3102A-18-1P	A	Ground
	B	Spare
	C	Ready Indication
	D	28 V DC input
	E	Ground
	F	Load
	G	Bleed
	H	Release
	I	Swing and feed
	J	Spare

Source: https://assist.dla.mil Downloaded: 2026-10-17T08:07Z
Check the source to verify that this is the current version before use.

Receptacle Type	Pin Designation	Function
MS3102A-20-278	A	Ground
	B	25 vDC to remote control
	C	Ready indication
	D	Spare
	E	Muzzle Velocity control
	F	Load
	G	Bleed
	H	Release
	I	Swing and feed
	J	Muzzle Velocity control
	K	Spare
	L	Spare
	M	Spare
	N	Spare

3.6.3 Remote Control PEK-23/A47U-1

3.6.3.1 Function - The unit shall provide for remote control of the launcher.

3.6.3.2 Form Factor - The envelope dimensions of the remote control shall be in accordance with EX-403.

*3.6.3.3 Weight - The weight of the remote control shall not exceed 3.5 pounds.

3.6.3.4 Contents - The remote control shall contain the following assemblies, sub-assemblies, circuits, etc.

3.6.3.4.1 Mounting - The mounting shall be in accordance with EI-403.

* 3.6.3.4.2 Face Plate - The face plate shall mount devices necessary for operation of the equipment and shall have suitable legends engraved thereon.

3.6.3.4.3 Chassis - The chassis shall have provisions for mounting the parts which cannot be attached to the face of the unit.

* 3.6.3.4.4 Power Switch - The power switch shall be a rotary selector switch. In the "OFF" position the switch shall interrupt electrical power from the aircraft to the remote control & the "STAND-BY" position electrical power shall be supplied to those parts requiring reheating prior to satisfactory operation. In the “OPERATE" position electrical power shall be supplied to operate the remote control and the launcher.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:07Z
Check the source to verify that this is the current version before use.

* 3.6.3.4.5 Ground Speed Indicator - The ground speed indicator shall indicate the muzzle velocity at which rounds are to be ejected. The indicator shall be a digital veeder root type instrument calibrated in knots of ground speed in one knot increments and shall have integral lighting.

* 3.6.3.4.6 Ground Speed Selector - A standard knob and crank shall be provided to operate the muzzle velocity control circuits and ground speed indicator when the unit is in remote operation

3.6.3.4.7 Ready Indicator - An indicator light shall be provided which shall be lighted when the equipment is ready in all respects for ejecting rounds with a release signal from the remote control.

3.6.3.4.8 Release Switch - The release switch shall be a guarded push-button type switch. Pressing the switch shall apply electrical power from the power switch shall apply to the firing valve.

* 3.6.3.4.9 Cover Box - The cover box shall be in accordance with 3.6.2.4.5.

*3.6.3.5 Electrical Connections - Connections to external circuits shall be provided by an MS3102A-20-27P connector as follows:

Pin Designation	Function
A	Ground
B	28 V DC from local control
C	Ready indication
D	Spare
E	Muzzle Velocity Control
F	Load.
G	Bleed
H	Release
I	swing and feed
J	Muzzle velocity control
K	Auxiliary release input
L	Auxiliary release power output
M	Spare
N	Spare

*3.6.4 Air Reservoir TMK-60\A47U-1 - The air reservoir shall be a Type 30-58, Class B. nonshatterable steel air reservoir in accordance with Specification MIL-R-8573. The capacity of the air reservoir shall be 50 cubic inches minimum.

3.6.5 Magazine MAK-22A/A47U-1

*3.6.5.1 Function - The magazine shall be capable of stowing at least 20 rounds so that they are accesible to the feed-mechanism of the launcher when the magazine is installed on the launcher.

Source: https://assist.dla.mil Downloaded: 2026-10-17T08:07Z
Check the source to verify that this is the current version before use.

* 3.6.5.2 Form Factor - The envelope dimensions of the magazine shall not exceed 20 inches long by 9 inches wide by 33 inches high.

3.7 Rounds - The equipment shall operate with the following rounds which shall not be supplied as part of the equipment.

Round	Size (inches)	Weight (pounds)
MK 7 Mod 2 Marine Marker	2.920 ±0.025D by 18.50 ± 0. 045L	3.25
MK 25 Mod O Marine Marker	2.970 ±0.010D by 18.45 ±0.098L	4

4. QUALITY ASSURANCE PROVISIONS

* 4.1 General - The inspection and testing of equipment covered by this specification shall be performed in accordance with the quality assurance provisions of Specification MIL-L-22769, unless otherwise specified herein.

* 4.2 Design Approval Tests do not apply.

* 4.3 Preproduction Tests - Preproduction tests are required only when procurement is made from a source which has not previously furnished equipment in accordance with this specification. Preproduction tests shall consist of:

(1) Unasterisked tests listed in Table II of Specification MIL-L-22769.

(2) Interference tests in accordance with Specification MIL-I-6181.

(3) Proof and burst pressure tests in accordance with Specification MIL-P-5518.

(4) Acceleration testing in accordance with Specification MIL-E-5272, Procedure X, so that the magnitude and direction of the resultant loads derived in accordance with Specification MIL-A-8591 are applied to the equipment.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-7936.

6. NOTES

*6.1 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 V DC and 115 ± 1.0 V AC, and pneumatic pressures of 3000 ± 50 PSIG. These data are to be used in testing the equipment at installations points for compliance th minimum acceptable standard of performance.

6.2 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply

(1) Contract - The contract shall have precedence over any specification.

(2) This specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, Or from subsidiary specification where applicable, shall be specifically approved in writing by the procuring activity.

Source: https://assist.dla.mil - Downloaded: 2026-10-17T08:07Z
Check the source to verify that this is the current version before use.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

*6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design performance ease of maintenance or reliability will result from the use of materials, parts, and processes other. than those specified in Specification MIL-L-22769, it is desired that their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by-complete supporting information.

6.4 The parentheses (*), when used in the type designation, may be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

*6.5 Ordering Data - Procurement documents shall specify:

(1) The title and number of this specification.

(2) The level of shipment desired. When level B is desired, the contract should specify the details of preservation and packaging (See 5.1).

(3) Design data in accordance with Specification MIL-D-18300.

(4) New and revised Engineering Drawings, Associated Lists, and Documents in accordance with Naval Weapons Requirements WR-12.

(5) Handbook Material.

6.6 An asterisk preceding a paragraph number denotes a change in that paragraph from the "C" version of this specification.

NOTICE - When Government drawings, speclfications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:07Z
Check the source to verify that this is the current version before use.